LAPORAN TUGAS

MATA KULIAH WORKSHOP DEVELOPER OPERATIONAL

RIO WILDAN R.A

NIM E41232146

Golongan E

PROGRAM STUDI TEKNIK INFORMATIKA

JURUSAN TEKNOLOGI INFORMASI

POLITEKNIK NEGERI JEMBER

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BAB 1 PEMBAHASAN

1.1 CI/CD

CI/CD adalah metode untuk mengirimkan aplikasi kepada pengguna secara terus-menerus dengan menggunakan otomatisasi. Terdiri dari dua bagian utama:

1) Continuous Integration (CI): Setiap kali developer mengubah kode, kode tersebut langsung digabungkan ke pusat penyimpanan

(repository), lalu diuji secara otomatis. Tujuannya agar kesalahan (bug) ditemukan lebih cepat.

2) Continuous Delivery/Deployment (CD): Setelah lulus uji, kode tersebut otomatis disiapkan untuk dirilis (Delivery) atau bahkan langsung diluncurkan ke server produksi (Deployment) tanpa campur tangan manusia.

1.2 Jenkins

Jenkins adalah tools *open-source* yang menjalankan proses CI/CD tersebut.

Jenkins bertugas :

1) Mengambil kode terbaru dari GitHub/GitLab.
2) Menjalankan perintah untuk membangun (*build*) aplikasi.
3) Menjalankan tes otomatis.
4) Mengirimkan hasilnya ke server tujuan

1.3 Jenkins Installation (WSL)

a. Masuk ke WSL

```
root@LAPTOP-K15MPFR2: /h

Microsoft Windows [Version 10.0.22631.6199]
(c) Microsoft Corporation. All rights reserved.

C:\Users\RIO>wsl
riowildan@LAPTOP-K15MPFR2:/mnt/c/Users/RIO$ sudo su
[sudo] password for riowildan:
root@LAPTOP-K15MPFR2:/mnt/c/Users/RIO# cd /home/riowildan
root@LAPTOP-K15MPFR2:/home/riowildan# sudo apt install openjdk-17-jdk -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openjdk-17-jdk is already the newest version (17.0.18+8-1~24.04.1).
0 upgraded, 0 newly installed, 0 to remove and 5 not upgraded.
root@LAPTOP-K15MPFR2:/home/riowildan#
```

Buka terminal dan masuk ke mode root/user utama.

b. Install Java JDK

Java JDK dibutuhkan untuk menjalankan Jenkins,disinsaya menggunakan versi Java JDK 17.

```
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Ign:3 https://pkg.jenkins.io/debian-stable binary/ InRelease
Hit:4 https://pkg.jenkins.io/debian-stable binary/ Release
Hit:6 https://download.docker.com/linux/ubuntu noble InRelease
Get:7 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:8 https://apt.releases.hashicorp.com noble InRelease [12.9 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.1 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:18 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7312 B]
Get:19 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:21 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 1070 kB in 23s (46.7 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
19 packages can be upgraded. Run 'apt list --upgradable' to see them.
W: https://apt.releases.hashicorp.com/dists/noble/InRelease: Key is stored in legacy trusted.gpg keyring (/etc/apt/trusted.gpg),
see the DEPRECATION section in apt-key(8) for details.
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# sudo apt install openjdk-17-jdk -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openjdk-17-jdk is already the newest version (17.0.18+8-1~24.04.1).
0 upgraded, 0 newly installed, 0 to remove and 19 not upgraded.
root@LAPTOP-PLBPPBOU:/home/devi_zahra_safrida# java -version
openjdk version "17.0.18" 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-124.04.1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-124.04.1, mixed mode, sharing)
```

c. Tambahkan Jenkins Repository Key

```
root@LAPTOP-K15MPFR2:/home/riowildan# sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
  https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
--2026-03-13 05:19:30--  https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.46.133, 2a04:4e42:7b::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|146.75.46.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1680 (1.6K) [application/octet-stream]
Saving to: '/etc/apt/keyrings/jenkins-keyring.asc'

/etc/apt/keyrings/jenkins-key 100%[===================================================>]   1.64K  --.-KB/s    in 0s

2026-03-13 05:19:31 (19.2 MB/s) - '/etc/apt/keyrings/jenkins-keyring.asc' saved [1680/1680]
```

d. Tambahkan Jenkins Repository

```
root@LAPTOP-K15MPFR2:/home/riowildan# echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc]" \
  https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
  /etc/apt/sources.list.d/jenkins.list > /dev/null
```

e. Install Jenkins

```
root@LAPTOP-K15MPFR2:/home/riowildan# sudo apt update
Get:1 https://apt.releases.hashicorp.com noble InRelease [12.9 kB]
Ign:2 https://pkg.jenkins.io/debian-stable binary/ InRelease
Hit:3 https://pkg.jenkins.io/debian-stable binary/ Release
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:5 http://archive.ubuntu.com/ubuntu noble InRelease
Get:7 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1512 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1809 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [243 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [976 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [219 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.7 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:20 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [33.0 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:24 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7312 B]
Get:25 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [30.8 kB]
Get:26 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:27 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:28 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 5909 kB in 6s (973 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
40 packages can be upgraded. Run 'apt list --upgradable' to see them.
W: https://apt.releases.hashicorp.com/dists/noble/InRelease: Key is stored in legacy trusted.gpg keyring (/etc/apt/trusted.gpg), see the DEPRECATION section in apt-key(8) for details.
root@LAPTOP-K15MPFR2:/home/riowildan# sudo apt install jenkins -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
jenkins is already the newest version (2.541.2).
0 upgraded, 0 newly installed, 0 to remove and 40 not upgraded.
root@LAPTOP-K15MPFR2:/home/riowildan#
```

Setelah instalasi, jalankan Jenkins dan aktifkan agar otomatis berjalan saat startup:

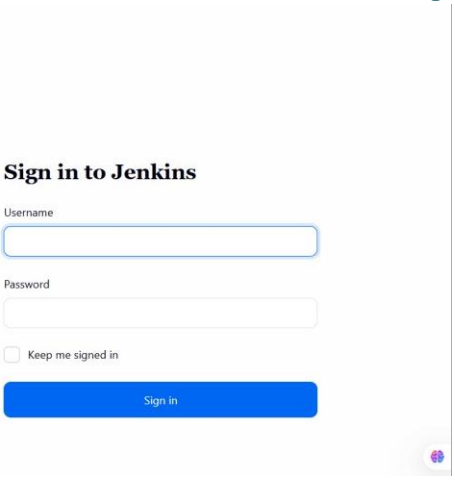
```
root@LAPTOP-K15MPFR2:/home/riowildan# sudo systemctl start jenkins
root@LAPTOP-K15MPFR2:/home/riowildan# sudo systemctl enable jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
root@LAPTOP-K15MPFR2:/home/riowildan# sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-03-12 22:52:40 WIB; 7h ago
   Main PID: 171 (java)
      Tasks: 59 (limit: 4332)
     Memory: 887.3M (peak: 1.0G)
        CPU: 3min 21.388s
     CGroup: /system.slice/jenkins.service
             └─171 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Mar 12 22:52:40 LAPTOP-K15MPFR2 jenkins[171]: 2026-03-12 15:52:40.804+0000 [id=66]	INFO	jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
Mar 12 22:52:40 LAPTOP-K15MPFR2 jenkins[171]: 2026-03-12 15:52:40.825+0000 [id=66]	INFO	jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
Mar 12 22:52:40 LAPTOP-K15MPFR2 jenkins[171]: 2026-03-12 15:52:40.848+0000 [id=88]	INFO	hudson.util.Retrier#start: Attempt #1 to do the action check updates server
Mar 12 22:52:40 LAPTOP-K15MPFR2 jenkins[171]: 2026-03-12 15:52:40.897+0000 [id=66]	INFO	jenkins.InitReactorRunner$1#onAttained: Completed initialization
Mar 12 22:52:40 LAPTOP-K15MPFR2 jenkins[171]: 2026-03-12 15:52:40.944+0000 [id=33]	INFO	hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
Mar 12 22:52:40 LAPTOP-K15MPFR2 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Mar 12 22:52:57 LAPTOP-K15MPFR2 jenkins[171]: 2026-03-12 15:52:57.673+0000 [id=88]	INFO	h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenInstaller
Mar 12 22:52:59 LAPTOP-K15MPFR2 jenkins[171]: 2026-03-12 15:52:59.751+0000 [id=88]	INFO	h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Ant.AntInstaller
Mar 12 22:53:20 LAPTOP-K15MPFR2 jenkins[171]: 2026-03-12 15:53:20.671+0000 [id=88]	WARNING	h.m.DownloadService$Downloadable#updateNow: No tool installer metadata found for hudson.plugins.gradle.
Mar 12 22:53:20 LAPTOP-K15MPFR2 jenkins[171]: 2026-03-12 15:53:20.675+0000 [id=88]	INFO	hudson.util.Retrier#start: Performed the action check updates server successfully at the attempt #1
lines 1-20/20 (END)
```

f. Cek apakah Jenkins berjalan

```
riowildan@LAPTOP-K15MPFR2:/mnt/c/Users/RIO$ sudo su
[sudo] password for riowildan:
root@LAPTOP-K15MPFR2:/mnt/c/Users/RIO# sudo lsof -i :8080
COMMAND PID    USER   FD   TYPE DEVICE SIZE/OFF NODE NAME
java    171 jenkins    8u  IPv6   1403      0t0  TCP *:http-alt (LISTEN)
root@LAPTOP-K15MPFR2:/mnt/c/Users/RIO#
```

g. Buka Jenkins Dashboard

Ketika port 8080 bisa diakses melalui http://localhost:8080 berarti



Jenkins berhasil dijalankan.

Kemudian ambil password awal Jenkins dengan perintah :

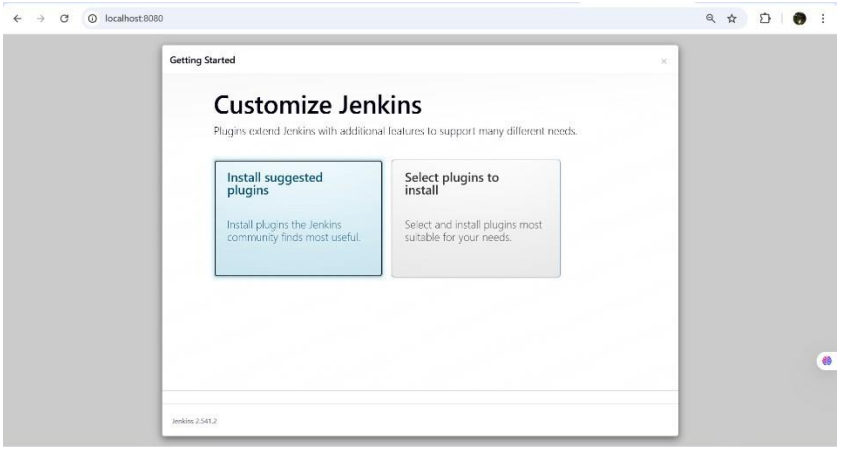
```
root@LAPTOP-K15MPFR2:/home/riowildan# sudo cat /var/lib/jenkins/secrets/initialAdminPassword
4bd2b05eecb4ef99888eb51b88514c7
```

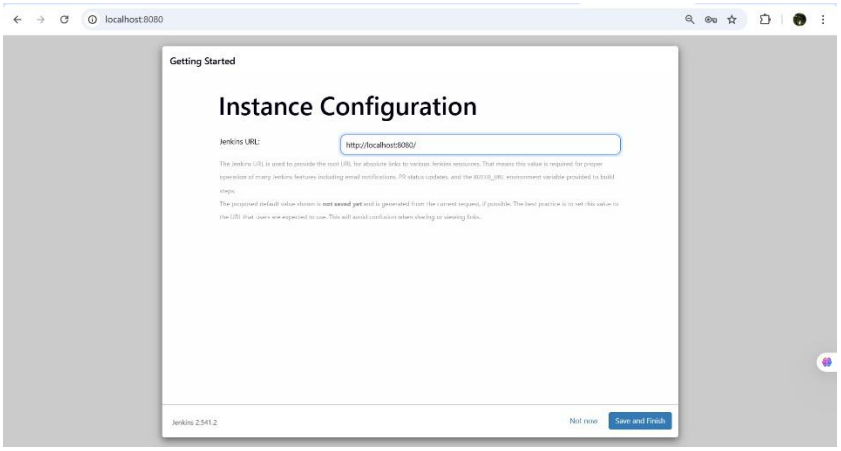
Password ini digunakan pada saat di halaman login Jenkins.

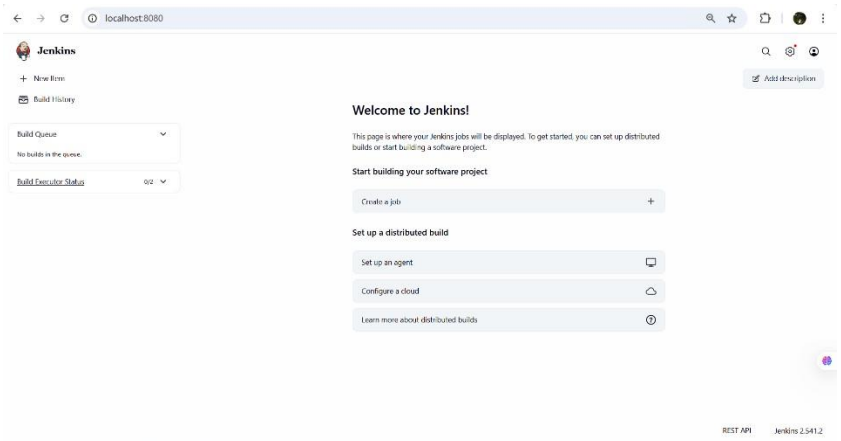
h. Install Jenkins Plugins

Pada bagian ini, pilih “Install Suggested Plugins”.

Setelah selesai, Jenkins akan masuk ke Dashboard.

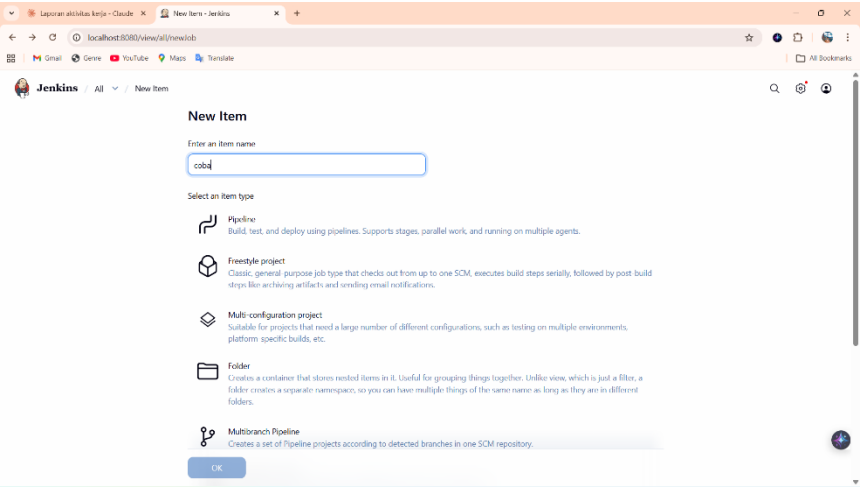




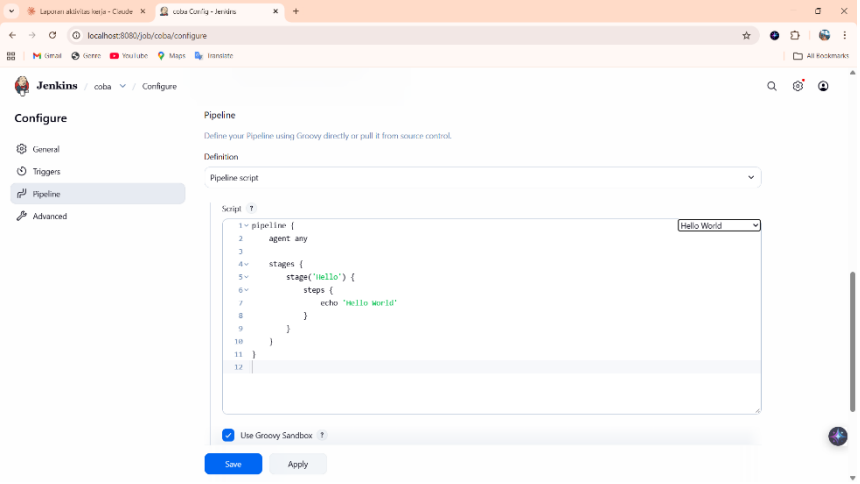


i. Test Jenkins Pipeline

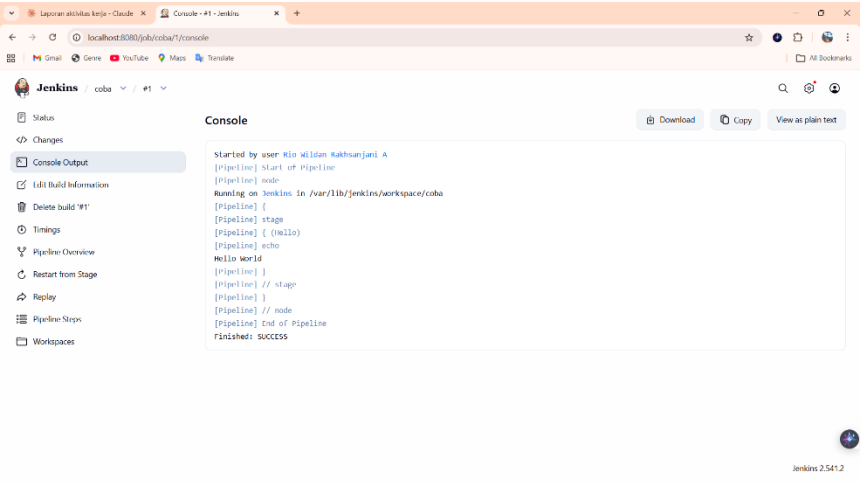
Disini kita test Pipeline dengan membuat job sederhana. Pilih ‘New Item’ dan Beri nama ‘coba’, dan pilih type pipeline.



Lalu, pilih definition ‘Pipeline Script” dan pada script pilih “Hello Word”, dan save.



Kemudian, klik ‘Build Now’, lihat hasilnya di ‘Console Ouput’, apabila muncul ‘Finished: SUCCESS’, maka dinyatakan telah berhasil.



1.4 Laravel Project Setup

a. Buat Folder Project

Masuk ke cd C:\Users\USERNAME\Documents, lalu buat direktori website dan Laravel_jenkins.

```
C:\Users\RIO>cd Documents

C:\Users\RIO\Documents>mkdir website

C:\Users\RIO\Documents>cd website

C:\Users\RIO\Documents\website>mkdir laravel_jenkins
```

b. Install Laravel

Laravel ini nantinya digunakan sebagai project yang akan dijalankan.

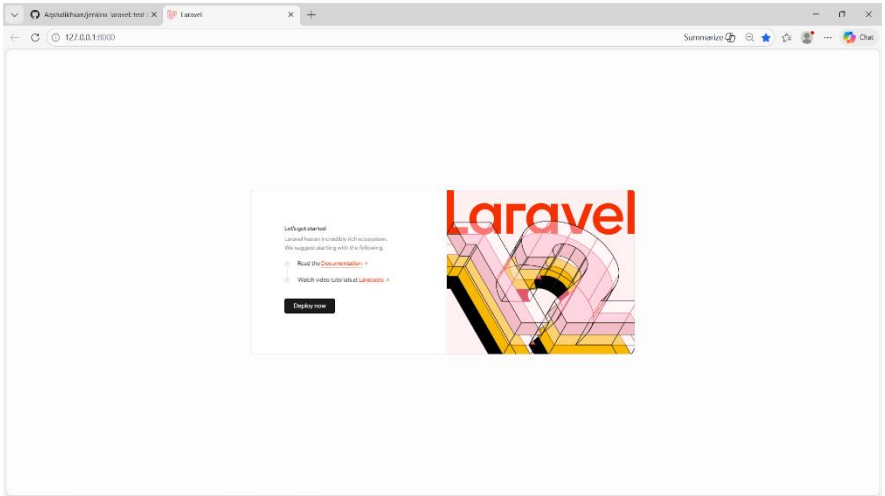
```
C:\Users\RIO\Documents\website>composer create-project laravel/laravel laravel-ci2
Creating a "laravel/laravel" project at "./laravel-ci2"
Installing laravel/laravel (v12.12.1)
  - Downloading laravel/laravel (v12.12.1)
  - Installing laravel/laravel (v12.12.1): Extracting archive
Created project in C:\Users\RIO\Documents\website\laravel-ci2
```

Ini saya bikin nama project laravel-ci2 karena saya lupa tidak ss dari awal, jadi saya membuat lagi buat dokumentasi. File asli saya yang sudah dibuat namanya LARAVEL-CI

c. Setup Environment dan jalankan Laravel

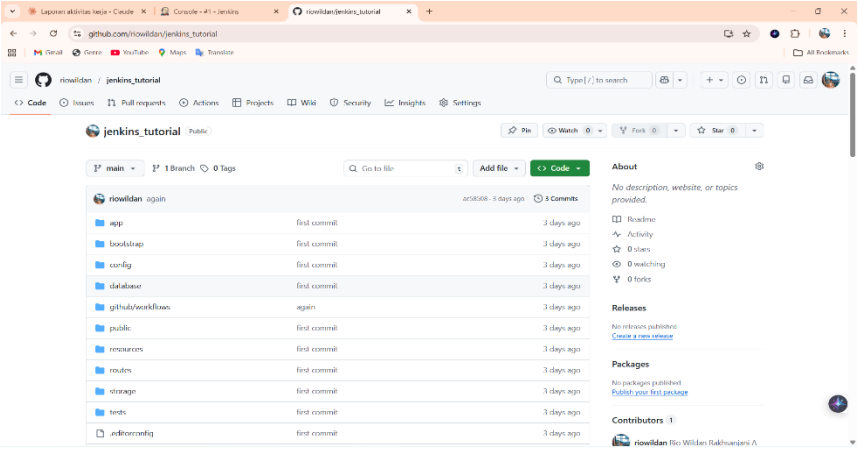
```
> @php artisan package:discover --ansi

  INFO  Discovering packages.

  laravel/pail ..................................................... DONE
  laravel/sail ..................................................... DONE
  laravel/tinker ................................................... DONE
  nesbot/carbon .................................................... DONE
  nunomaduro/collision ............................................. DONE
  nunomaduro/termwind .............................................. DONE

81 packages you are using are looking for funding.
Use the `composer fund` command to find out more!
> @php artisan vendor:publish --tag=laravel-assets --ansi --force

  INFO  No publishable resources for tag [laravel-assets].

No security vulnerability advisories found.
> @php artisan key:generate --ansi

  INFO  Application key set successfully.

> @php -r "file_exists('database/database.sqlite') || touch('database/database.sqlite');"
> @php artisan migrate --graceful --ansi

  INFO  Preparing database.

  Creating migration table ......................................... 12.50ms DONE

  INFO  Running migrations.
```

Lalu akses di browser melalui http://127.0.0.1:8000

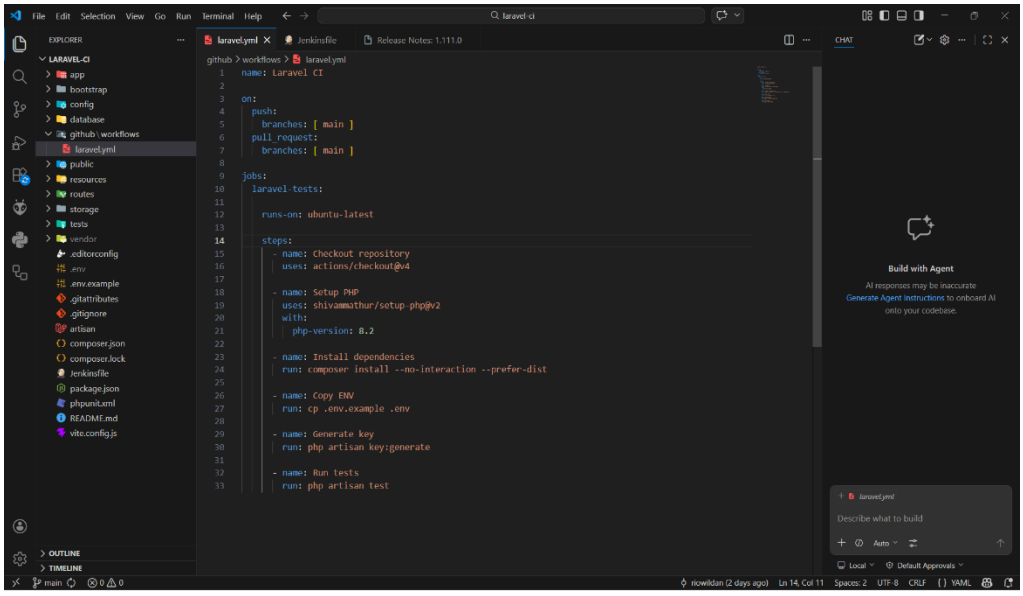


1.1 CI/CD using GitHub Actions

Setelah berhasil membuat project Laravel, selanjutnya push project ke GitHub. Create repository dengan nama ‘jenkins_laravel’.



Buka project Laravel bisa melalui VSCode. Lalu buat folder workflow ‘.github/workflows’, lalu buat file didalamnya ‘laravel.yml’. Setelah itu, push lagi ke github.



1.5 CI/CD using Jenkins

a. Install Required Jenkins Plugins

Masuk ke Manage Jenkins, lalu pilih Plugins. Install beberapa plugin berikut :

- Git plugin
- GitHub Integration
- Credentials Binding
- Pipeline
- Pipeline: Stage View

Selanjutnya, install tools php dan composer.

```
root@LAPTOP-K15MPFR2:/home/riowildan# sudo apt install php-cli php-json php-zip php-mbstring unzip curl -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
php-cli is already the newest version (2:8.3+93ubuntu2).
php-cli set to manually installed.
php-zip is already the newest version (2:8.3+93ubuntu2).
php-zip set to manually installed.
php-mbstring is already the newest version (2:8.3+93ubuntu2).
php-mbstring set to manually installed.
```

```
root@LAPTOP-K15MPFR2:/home/riowildan# curl -sS https://getcomposer.org/installer | php
All settings correct for using Composer
Downloading...

Composer (version 2.9.5) successfully installed to: /home/riowildan/composer.phar
Use it: php composer.phar

root@LAPTOP-K15MPFR2:/home/riowildan# sudo mv composer.phar /usr/local/bin/composer
root@LAPTOP-K15MPFR2:/home/riowildan# composer -v
Do not run Composer as root/super user! See https://getcomposer.org/root for details
Continue as root/super user [yes]? yes
```

b. Install PHP & Composer on Jenkins (WSL)

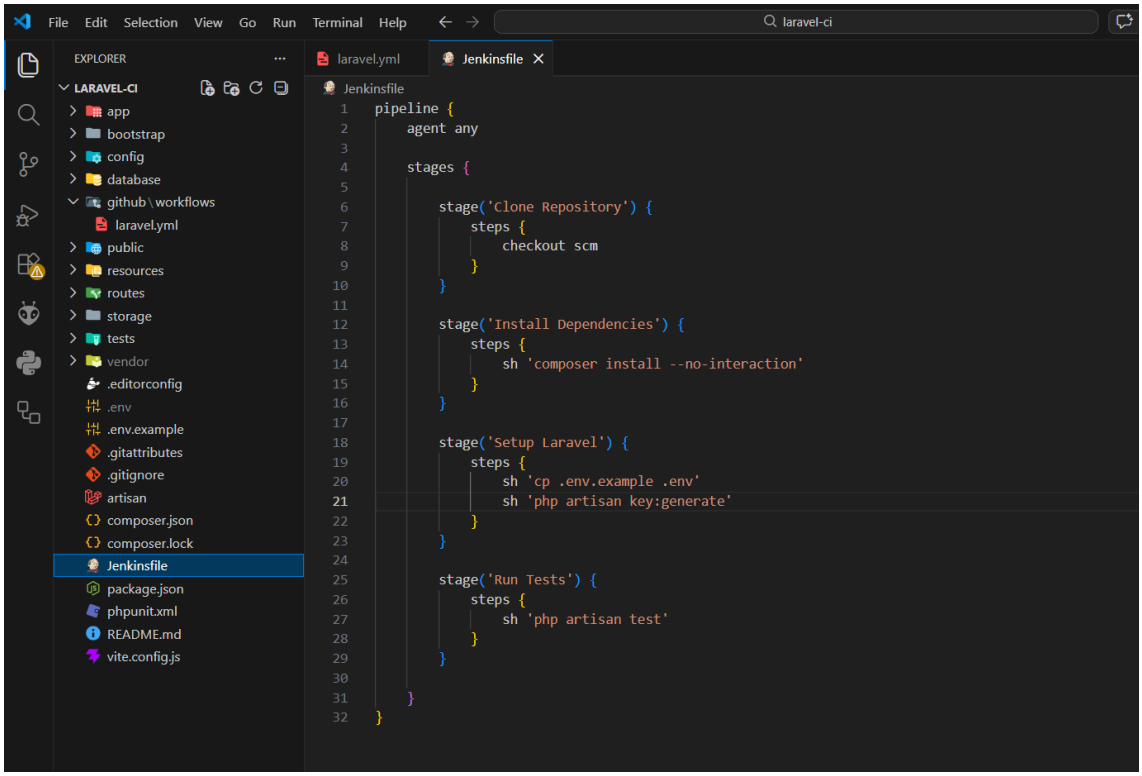
Selanjutnya, install php dan composer di WSL

```
root@LAPTOP-K15MPFR2:/home/riowildan# sudo apt install php-cli php-json php-zip php-mbstring unzip curl -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
php-cli is already the newest version (2:8.3+93ubuntu2).
php-cli set to manually installed.
php-zip is already the newest version (2:8.3+93ubuntu2).
php-zip set to manually installed.
php-mbstring is already the newest version (2:8.3+93ubuntu2).
php-mbstring set to manually installed.
unzip is already the newest version (6.0-28ubuntu4.1).
curl is already the newest version (8.5.0-2ubuntu10.7).
The following NEW packages will be installed:
  php-json
0 upgraded, 1 newly installed, 0 to remove and 5 not upgraded.
Need to get 1838 B of archives.
After this operation, 11.3 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 php-json all 2:8.3+93ubuntu2 [1838 B]
Fetched 1838 B in 1s (3466 B/s)
Selecting previously unselected package php-json.
(Reading database ... 44921 files and directories currently installed.)
```

```
root@LAPTOP-K15MPFR2:/home/riowildan# sudo apt install composer -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
composer is already the newest version (2.7.1-2).
0 upgraded, 0 newly installed, 0 to remove and 5 not upgraded.
```

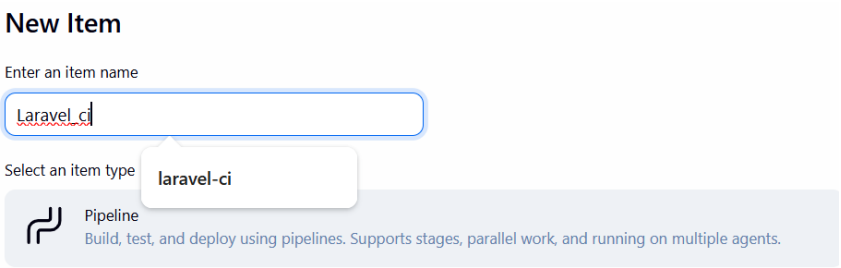
c. Jenkins Pipeline Script

Buat file di root project Laravel di VSCode dengan nama file 'Jenkinsfile'. Lalu push file ini ke GitHub.

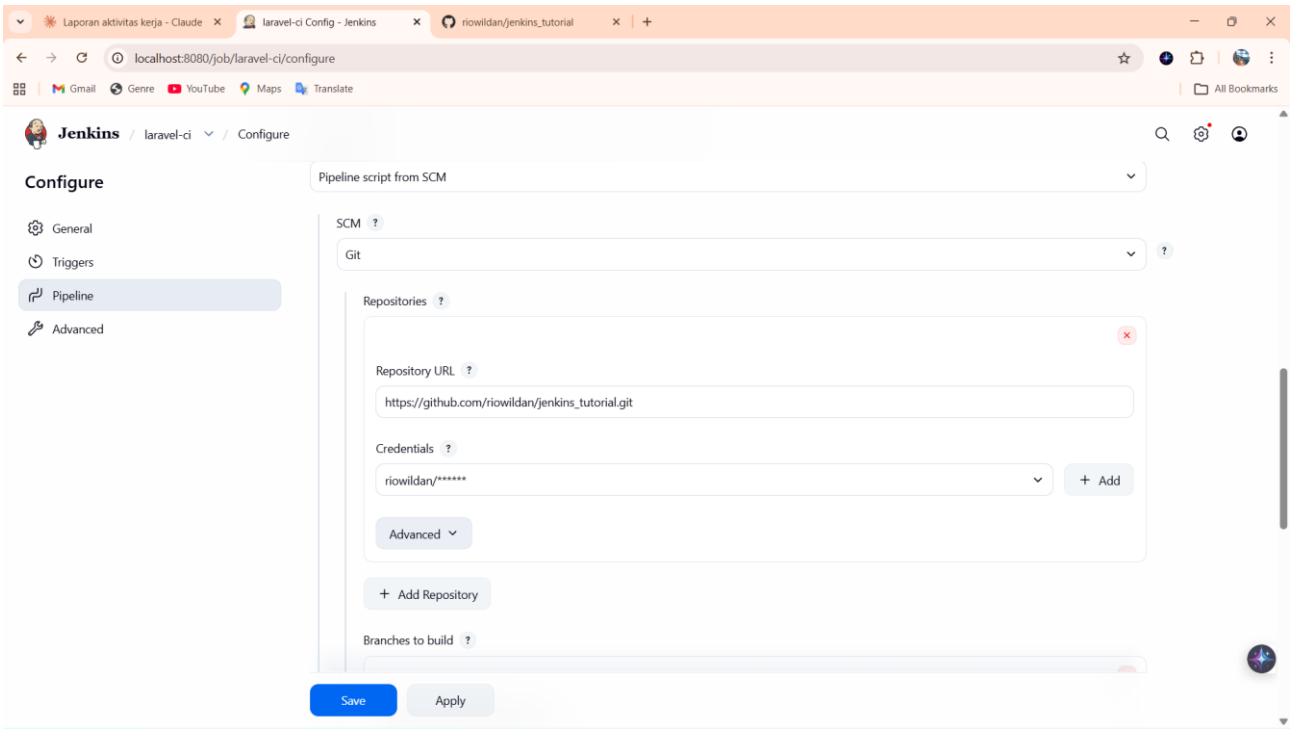


d. Jenkins Pipeline Configuration

Masuk ke halaman Jenkins, lalu pilih 'New Item', beri nama item 'laravel-ci', dan pilih tipe 'Pipeline'.



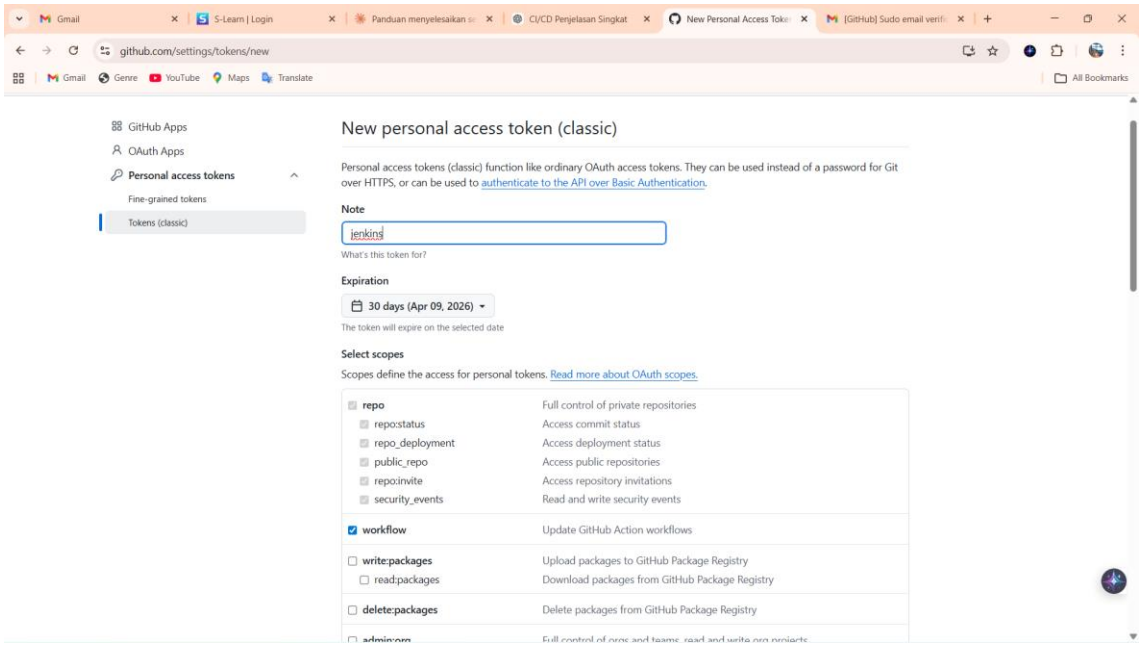
Atur configurasi Pipeline. Pada Definition pilih ‘Pipeline script from SCM’, pada SCM pilih ‘Git’, pada Repo masukkan link GitHub, dan pada Branch masukkan ‘*/main’.



e. GitHub Credential Setup

Untuk memasukkan repository GitHub pada Pipeline Configuration diperlukan token dari GitHub. Jadi bikin tokennya dahulu.

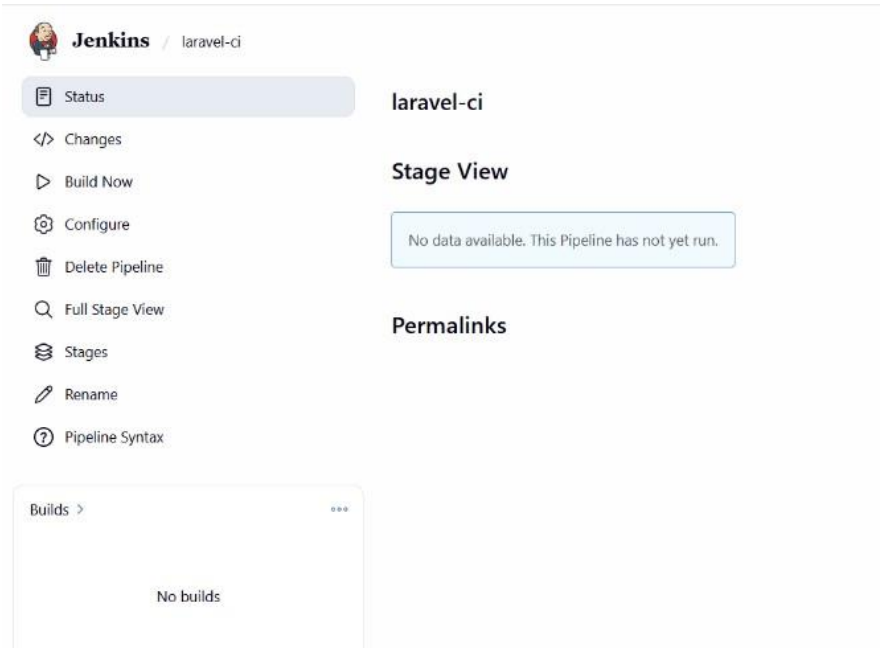
Untuk generate GitHub Toke, masuk ke pengaturan GitHub, pilih ‘Developer settings’, pilih ‘Personal access tokens’, dan pilih ‘Tokens (classic). Centang permission repo dan workflow



f. Run Jenkins Pipeline

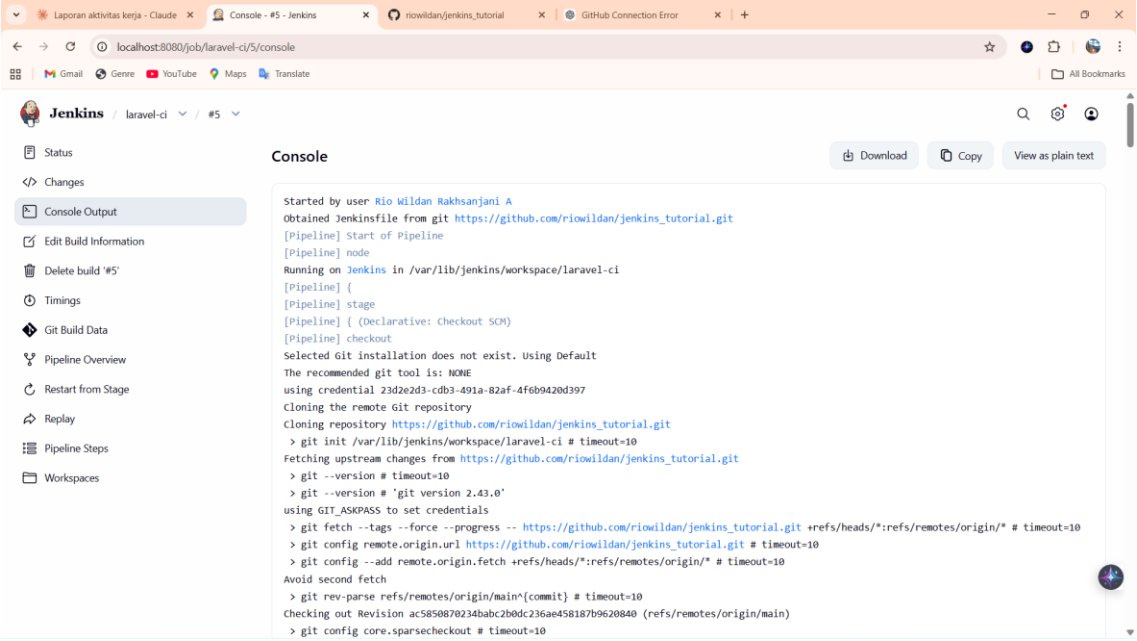
Selanjutnya yaitu menjalankan Jenkins Pipeline. Klik ‘Build Now’, maka akan memulai build pipeline. Pipeline akan menjalankan Clone Repository, Composer Install, Laravel Setup, dan Run Tests.

Jika berhasil, maka akan muncul Finished: SUCCESS di console.



1.1 Final CI/CD Workflow

Terlihat run Jenkins Pipeline



Alur kerja otomatisasi (Workflow), sebagai berikut :

1. Developer Push
   Proses dimulai ketika mengirimkan perubahan kode terbaru dari komputer lokal ke GitHub menggunakan perintah git push origin main.
2. GitHub Repository
   GitHub bertindak sebagai pusat penyimpanan kode (Single Source of Truth). Setiap perubahan yang masuk akan memicu webhook untuk menjalankan tahap berikutnya.
3. GitHub Actions CI
   Sebelum kode masuk ke Jenkins, GitHub Actions melakukan pengecekan awal di awan (Cloud).
   Fungsi: Menjalankan linter atau tes cepat pada lingkungan ubuntu-latest. Keuntungan: Mendeteksi error sintaksis dasar sebelum membebani server Jenkins lokal Anda.
4. Jenkins Pipeline

Setelah lolos dari GitHub Actions, Jenkins (yang berjalan di WSL) mengambil alih kendali. Jenkins akan menarik kode terbaru (Pull) dan mengeksekusi instruksi di dalam Jenkinsfile.

5. Build + Test Laravel Application

   Ini adalah tahap pembuktian di lingkungan Environment yang sesungguhnya:

   - Install: Menjalankan composer install (mengunduh vendor).
   - Setup: Mengatur .env dan key:generate.
   - Test: Menjalankan php artisan test untuk memastikan fitur aplikasi tidak rusak setelah ada kode baru.